Closed Nucleus Theory: A Research Programme in Reproduction Physics

Ruster

June 2026

Abstract

Contemporary fundamental physics faces a set of difficulties that share a structural character and may involve ontological issues: the vacuum catastrophe ($\sim 10^{120}$ orders of magnitude discrepancy), the dark matter puzzle (decades of null WIMP detection results), the unexplained equivalence of inertial and gravitational mass, and the absence of a consensus quantum gravity framework. These problems share a common feature—they all concern the origin of mass, gravity, and spacetime structure, yet the mainstream paradigm treats them as *given* (mass as an intrinsic property, spacetime as a background stage) rather than *generated*.

This paper presents Closed Nucleus Theory (CNT) as a specific construction within a broader paradigm here termed *Reproduction Physics*. The paradigm's starting point is an ontological inversion derived from general relativity itself: curvature is not caused by mass-energy but is ontologically prior—mass is the emergent readout of energy quanta bound within geometric structures. From this curvature-first ontology, the persistent existence of massive structures demands a maintenance mechanism: this is reproduction, rigorously defined as *producing again the conditions of maintenance*, with no additional requirements. CNT instantiates this paradigm through a strong construction that adds proliferation, fission, irreducibility, and historical dependence, yielding a framework in which: (1) mass is not a source but an emergent quantity—the readout of energy quanta absorbed at closed nucleus boundaries; (2) gravity is a force—but fundamentally different from the three gauge forces, an ontological response in which spacetime curvature responds directly to the quantized stress-energy tensor $T_{\mu\nu} = \sum_i h\nu_i u_\mu^{(i)} u_\nu^{(i)}$ via the Einstein field equations; (3) closed nuclei are not particles placed in spacetime but local excitations *of* spacetime itself.

The mathematical structure of CNT is anchored in the regular 4-simplex as the minimal geometric realization of a closed nucleus, and in p-adic numbers as the necessary mathematics of reproduction physics—a valuable theoretical system should grow its needed mathematics from within: the chain from reproduction counting through prime irreducibility to the projective limit ($\mathbb{Z}_p = \varprojlim \mathbb{Z}/p^n\mathbb{Z}$) points to p-adic numbers as the necessary direction, but their specific form—standard p-adic numbers, compositional p-adic numbers, or a more general structure—remains to be explored. The framework is organized as a six-equation mathematical skeleton tracing the causal chain from microscopic geodesic motion to holographic mass emergence. The mathematical requirements this skeleton imposes—the Bekenstein bound as a saturation limit, gauge groups as distinct structural layers of the same reproduction structure, and the gravitational constant G as the energy quantum–

closed nucleus binding coupling constant—delineate the research frontier of the programme.

This is a *research programme*—an ongoing, disciplined, falsifiable theoretical construction. All gaps are explicitly labeled and graded by severity. This paper is an invitation to the physics community to examine, criticize, and—should the framework be judged to have merit—contribute to its development.

Keywords: Reproduction Physics, Closed Nucleus Theory, loop quantum gravity, spin foam models, ontological inversion, mass emergence, p-adic numbers, holographic principle, quantum gravity, spacetime excitation, projective limit, dark matter, research programme

Subject classification: Theoretical Physics, Quantum Gravity, Loop Quantum Gravity, Spin Foam Models, Foundations of Physics, Cosmology, Emergent Gravity

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1 Introduction

1.1 The Predicament of Contemporary Physics: A Possible Ontological Root

Contemporary fundamental physics is operationally extraordinarily successful. The Standard Model of particle physics, together with the ΛCDM cosmological model, explains an extraordinarily wide range of phenomena with a relatively small number of free parameters. Yet this success masks a deeper predicament: the theory explains neither *why* its fundamental parameters take the values they do, nor *what* several of its core entities actually are.

This predicament may have an ontological rather than a merely technical character. It is not a matter of missing higher-order corrections or insufficient computational power, but of the framework's fundamental assumptions about what exists and how. Specifically:

(1) **The vacuum catastrophe.** Quantum field theory predicts a vacuum energy density roughly 10^{120} times larger than the observed value. This is not a calculation error—it is an indication that the straightforward assumption "energy produces gravity" may require modification when applied to quantum vacuum fluctuations.

(2) **The dark matter puzzle.** The ΛCDM model requires a non-baryonic matter component constituting approximately 27% of the cosmic energy budget [3]. The leading particle candidate—Weakly Interacting Massive Particles (WIMPs)—has been the target of direct detection experiments (XENON1T, LZ, PandaX) and collider searches (LHC) for decades, all yielding null results [1, 2]. The WIMP parameter space has been systematically excluded to the point of a naturalness crisis.

(3) **The equivalence principle.** The equality of inertial and gravitational mass is a postulate of general relativity, not a derived result. Why the quantity that resists acceleration equals the quantity that generates gravity remains unexplained.

(4) **The Higgs mechanism.** Introducing a fundamental scalar field brings with it the hierarchy problem and the naturalness problem. Without fine-tuning, the Higgs boson mass is unstable under radiative corrections.

(5) **Quantum gravity.** String theory, loop quantum gravity (LQG) [7, 8], causal set theory, and asymptotic safety are competing approaches with no experimental consensus and no shared conceptual framework. Among these, LQG—and its covariant spin-foam formulation in particular—bears a natural structural resonance with Reproduction Physics: both treat spacetime geometry as the fundamental ontological entity, both reject a fixed background metric at the quantum level, and both describe physical evolution in terms of discrete geometric excitations. This resonance is developed in §3.3 and §4.1.

The common feature of these problems is that they all concern the origin of mass, gravity, and spacetime structure. The mainstream paradigm treats them as *given*—mass is an intrinsic property of matter, spacetime is the background stage of physical evolution, gravity is spacetime's curvature response to mass-energy. The question CNT poses is: *What if this explanatory order is inverted?*

1.2 The Ontological Inversion

The core move of Reproduction Physics is not an arbitrary choice but a *structural derivation* from general relativity itself. The Einstein field equations

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \tag{1}$$

encode a profound ontological ambiguity: geometry (the left-hand side) determines how the stress-energy tensor moves, and the stress-energy tensor (the right-hand side) determines how geometry curves. The standard interpretation treats this as mutual determination—matter tells space how to curve, space tells matter how to move. But this relation is not symmetric at the ontological level. *What exists prior to what?*

CNT proposes that curvature—geometric structure—is ontologically prior. The stress-energy tensor is not an independent source that *causes* curvature; rather, curvature is the intrinsic state, and what we call "mass-energy" is the readout of how energy quanta are dynamically bound within geometric structure. This is the **weak ontological inversion**: it reframes the Einstein equations from a causal relation (mass causes curvature) to a constitutive relation (curvature binds energy; mass is the emergent readout of this binding).

This weak inversion has an immediate and natural consequence: it dissolves the vacuum catastrophe at the conceptual level. If mass is not a source but an emergent readout of bound energy quanta, then unbound vacuum fluctuations—however energetic—do not, in the straightforward sense, produce mass. The 10^{120} discrepancy is a symptom of treating vacuum energy as a source term in a framework where the ontological priority has been inverted. This does not *solve* the vacuum catastrophe but reframes it: the question is no longer "why does vacuum energy not gravitate?" but "under what geometric conditions do energy quanta become bound and thus contribute to mass?"

The weak inversion naturally elevates to the **strong ontological inversion**. If curvature is intrinsic and mass is the readout of bound energy, then the persistent existence of any massive structure requires a *maintenance* mechanism for energy-quantum binding. Binding is not a one-time event but a condition that must be continuously renewed. This is where reproduction enters: physical structures persist because they *reproduce their own conditions of maintenance*—the geometric conditions that bind energy quanta and thereby sustain mass. The strong ontology thus follows from the weak ontology through a chain of structural necessity:

Standard picture: mass (as source) → gravity (as force/curvature) → spacetime structure.
Weak inversion (curvature-priority): intrinsic spacetime curvature → geometric carrier of energy quanta → energy quanta bound within closed geometric regions → mass as emergent readout of bound energy.
Strong inversion (reproduction): intrinsic curvature drives energy quanta along geodesics → the geometric conditions of binding are *reproduction conditions* → energy quanta dynamically renew these conditions → mass sustained through reproduction → gravity as ontological curvature response.

This chain is not externally imposed. It is the structural consequence of taking curvature-first ontology seriously. The strong inversion additionally *inverts the principle of least action*: the traditional view treats the Einstein–Hilbert action $\delta \int (R +$

$\mathcal{L}_m)\sqrt{-g}\,d^4x = 0$ as the *cause* of dynamics—spacetime extremizes a functional. In the reproduction framework, the variational principle is a *consequence*: reproduction's intrinsic algebraic structure (idempotence, irreversibility, historical dependence) drives the dynamics, and the action principle is an effective description of this underlying process at the continuum level. The algebraic structure is the cause; the variational principle is the phenomenological signal.

This means that CNT is not a closed, self-consistent theory constructed from arbitrary axioms. It is a path of exploration that inherits the empirical success of general relativity while reinterpreting its ontological structure. The theory's validity rests entirely on whether the ontological inversion—the claim that curvature is ontologically prior to mass-energy—is correct. If this claim is false, the entire framework collapses. There are no secondary defenses, no tunable parameters that could rescue it. This single-point falsifiability is both discipline and strength.

In this inversion, mass is not a fundamental property of matter but an *emergent* quantity—the readout of energy quanta absorbed at the boundary of a localized spacetime excitation called a *closed nucleus*. Gravity is a force—but it is fundamentally different from the three gauge forces (electromagnetic, strong, weak). It is not mediated by gauge bosons. It is an *ontological response*: spacetime curvature responds directly to the quantized stress-energy tensor of bound energy quanta. The closed nucleus itself is not a particle *placed in* spacetime but a local excitation *of* spacetime—analogous to how a phonon is not a particle in a crystal lattice but a collective excitation of the lattice itself.

This inversion is not a refinement of existing theory. It is a shift of conceptual starting point. The question changes from "what are the laws governing given entities?" to "how do structures sustain themselves through reproduction?"

1.3 Philosophical Sources

The Reproduction Physics paradigm draws on three philosophical traditions without being reducible to any one of them:

- **Process philosophy**: existence requires process for evolution and persistence, not that existence is process. CNT provides what process philosophy has always lacked—a concrete, computationally tractable mathematical structure (p-adic numbers, projective limits, idempotent operators) that makes the *maintenance process* operable.
- **Historical materialism** (Marx): the concept of "reproduction" as the mechanism by which a structure maintains the conditions of its own existence. CNT extracts and generalizes this concept from its socio-economic context into an ontological principle: physical structures exist not because they are given, but because they *reproduce* their own conditions of maintenance. Historical materialism provides another crucial insight: structures themselves—whether electromagnetic structure, strong-force structure, weak-force structure, or particle structure—should not be treated as directly given, "free-lunch" entities. From a historical-materialist perspective, they must themselves possess an evolutionary process. The mathematical-physical edifice of physics has been extraordinarily successful at *statically characterizing* these structures, but the *historical evolutionary dimension* of the structures themselves has always been absent—and "historical" here does not mean the evolution of given structures under a time coordinate, but the evolution *of* the structures themselves. Introducing this historical dimension cannot be done through externally appended narratives; it must

enter from within the already-established mathematical-physical framework, through an ontological shift that maintains mathematical self-consistency.

- **Dialectics of nature—Aufhebung and mathematization:** Traditional dialectics, as a method of analysis, is suited to the analysis of society and human subjects, but cannot be mechanically transposed into the physical domain. Physics possesses its own methodology—mathematics—yet its historical evolutionary dimension has always been absent or incomplete. Engels' thesis—"nature is not a collection of completed things but a collection of processes"—is the correct point of departure. However, traditional dialectics of nature (including Engels' own formulations), having been incubated within the analysis of society and human subjects, carry social residues and human-subject residues, and these residues produce a fatal consequence: the impossibility of identifying a *mathematical entry point* for the dialectics of nature.

  To address this, CNT proposes an *Aufhebung* (sublation) of Engels' traditional dialectics of nature: eliminating the social residues and human-subject residues from the traditional dialectics of nature to establish the foundation for thoroughgoing mathematization, while retaining its dialectical core of historical hierarchical evolution, non-reducibility, and intrinsic driving force. The concrete physical form after Aufhebung is a three-link sequence—this is not the physical application of traditional dialectics, but the physicalized reconstruction of the dialectics of nature itself:

  (1) **Structural tension**—mathematical expression identified as modular arithmetic yielding non-zero remainders: any modular operation with non-zero remainder signals the presence of structural tension (the part-whole asymmetry irreducible in either direction); the mathematical tools are prime-number incommensurability and modular arithmetic;

  (2) **Quantity-to-quality transition**—the mathematical direction points to the p-adic number system (§4.2), where the accumulation of reproduction counts produces qualitative leaps through the p-adic stratified structure; their specific form (standard p-adic numbers, compositional p-adic numbers, etc.) is an active area of exploration;

  (3) **Hierarchical emergence**—mathematized as the renormalization group and meta-RG, where stability conditions of higher-level structures emerge from lower-level dynamics through the fixed-point structure of RG flows.

  The mathematization of structural tension is not yet complete ([**Gap 10**]), but the mathematical correspondences for quantity-to-quality (p-adic numbers) and hierarchical emergence (RG) are already firmly anchored within the framework.

This relationship is one of *ontological extraction*, not theoretical extension. The vocabulary of reproduction, proliferation, and historical dependence is borrowed from Marx; the structural-tension $\to$ quantity-to-quality $\to$ hierarchical emergence dialectic is a physicalized reconstruction of the Engels tradition after Aufhebung. All physical content is independently constructed from geometric, algebraic, and information-theoretic first principles.

In CNT's three-link dialectics of nature (structural tension $\to$ quantity-to-quality $\to$ hierarchical emergence), **remainder analysis** is the core of the structural-tension link. Non-zero modular remainder—the structural remainder produced when a higher level grasps the whole via the cumulative product P_k ($x \bmod P_k \neq 0$), and the coefficient remainder when a finite base cannot fully absorb the higher-level coefficient ($S_n \bmod p_n = a_{n,0} \neq 0$)—is not an accidental by-product of structure, but a *dynamical mechanism* of history. The remainder of each level becomes the seed for the next level's base generation ($p_{n+1} = S_n$), and the historical accumulation of remainders constitutes irreducible structural memory. This analysis differs from traditional dialectical descriptions of "contradiction"—it provides an explicit mathematical entry point for the first link (structural tension) of the dialectics of nature: modular arithmetic. The full mathematization of structural tension remains under construction ([**Gap 10**]), but remainder analysis already furnishes the core conceptual framework for this gap.

A natural corollary follows, though it is *not the only possible corollary*. At the intersection of historical materialism and the Aufhebung-elevated dialectics of nature, one natural reading is that gauge forces—electromagnetism, the strong force, the weak force—are not independently existing entities but distinct structural layers of the same reproduction structure, each corresponding to a different level of the same reproductive dynamics. Their origin requires historical-structural analysis—tracing the evolutionary process by which these layers emerged—but this is an analytical method, not an ontological claim about their present nature. This reading implies that the apparent independence of gauge forces is a surface phenomenon; their unity is rooted in the identity of the reproduction structure that manifests through them. This corollary acquires operational content in §3.6—the gauge group is precisely the intersection point of cyclology (time-structure analysis) and tomography (space-structure analysis).

1.4 Scope and Status of This Paper

This paper is a *research programme*, not a completed theory. It presents a framework under active construction, explicitly demarcating what has been achieved, what is conjectural, and what remains open. The goal is to enable the physics community to assess whether this framework merits serious engagement.

CNT is developed within the broader paradigm of Reproduction Physics (§3.3). Existing approaches to quantum gravity that are structurally compatible with this paradigm—most notably the covariant spin-foam formulation of loop quantum gravity (LQG), whose evolution amplitudes rest on the same geometric structures (4-simplices, $SU(2)$ representations) that CNT identifies as necessary consequences of the curvature-first ontology—may be understood as independent constructions within the same Reproduction Physics paradigm. This explicit identification serves to broaden the framework's detectability and to invite engagement from the LQG community.

The remainder of this paper is organized as follows. Section 2 establishes the ontological foundations: the definition of closed nuclei, the mechanism of mass emergence, the ontological status of gravity, and compatibility with special relativity. Section 3 presents the methodological framework: the distinction between dynamic and static analysis, the reproduction mechanism as the core of physical maintenance, the principles governing mathematical tool selection, and the cycle-tomography spacetime duality. Section 4 develops the mathematical structure: regular 4-simplex geometry, p-adic numbers as the necessary mathematics, gauge groups as structural layers, the proposed Historical Path Integral, and a working hypothesis on the network architecture of closed nuclei. Section 5 unfolds the mathematical requirements and structural constraints: the six-equation skeleton, the mathematical necessity of holographic boundaries, gauge groups as mathematical constraints, and the requirement that G be derived from energy quantum–closed nucleus binding dynamics. Section 6 explicitly catalogues the open problems and gaps in the framework. Section 7 concludes with a summary and an invitation to the community.

1.5 Notation and Terminology

Throughout this paper, the following terminological conventions are observed:

- **Reproduction**: rigorously defined as "producing again the conditions of maintenance." Its mathematical expression is the idempotent morphism $\mu : S \to S$, $\mu \circ \mu = \mu$.
- **Closed nucleus**: a local spacetime excitation, not a particle. Its minimal geometric realization is the regular 4-simplex.
- **Mass**: an emergent quantity, not a source. It is the readout of energy quanta absorbed at the closed nucleus boundary, converted via $E = mc^2$.
- **Gravity**: a force fundamentally different from the gauge forces—an ontological response. Spacetime curvature responds directly to the quantized stress-energy tensor via the Einstein field equations.
- **Energy quantum**: an energy quantum is simply an energy quantum. Energy quanta move along geodesics of the background spacetime, carrying quantized energy $h\nu$.
- **Existence**: the *factivity* of a closed nucleus as a spacetime excitation—it is the basal factuality within the curvature-first ontology. Existence is not generated by reproduction.
- **Maintenance / Reproduction**: the conditional process by which existence is *sustained*, rigorously defined as "producing again the conditions of maintenance." Existence is the foundation; evolution and maintenance are processes.

2 Ontological Foundations

2.1 Closed Nuclei as Spacetime Excitations

The fundamental ontological entity in CNT is the *closed nucleus*—a local excitation of spacetime itself. This concept follows directly from the curvature-first ontology: if curvature is intrinsic, then localized regions of non-trivial curvature—regions where geometric structure deviates from the flat Minkowski ground state—are the fundamental carriers of physical structure. The closed nucleus is precisely such a region. This is not metaphor; it is a precise ontological claim: just as a phonon is a quantized excitation of a crystal lattice, a closed nucleus is a quantized excitation of the spacetime manifold. The closed nucleus is not a particle *placed in* spacetime; it is a configuration *of* spacetime itself.

A closed nucleus is characterized by five necessary and sufficient conditions, which together formalize what it means for a geometric region to be a *persistent* spacetime excitation:

(C1) **Operational closure:** $\exists f : S \to S, f \circ f = f$. There exists an idempotent endomorphism on the structure, meaning the outcome of the operation is stable under iteration.

(C2) **Self-reference:** The structure distinguishes inside from outside, producing a minimal perspective—a boundary.

(C3) **Reproduction:** $\exists \mu : S \to S, \mu \circ \mu = \mu$. The reproduction morphism is idempotent: repeated reproduction does not alter the formal identity of the structure. This is the mathematical expression of "producing again the conditions of maintenance."

(C4) **Conditional debt:** $\exists \varepsilon : S \to S, \neg \mathrm{IsIso}(\varepsilon)$. There exists a non-isomorphic morphism—the closure condition is not permanently satisfied; it must be continuously renewed. This is the origin of the arrow of time.

(C5) **Historical sedimentation:** The selective stabilization of configurations through reproductive history, generating path-dependent structures.

These five criteria are not independent postulates but mutually reinforcing aspects of a single ontological entity. The idempotence of reproduction (C3) and the irreversibility of conditional debt (C4) jointly generate the tension between stable self-maintenance and irreversible change that drives the entire dynamics of the framework.

Ontological distinction between existence and maintenance. The above five criteria characterize the conditions for closed nuclei to *persist*, not the conditions for them to *exist*. The existence of a closed nucleus—as a local configuration of non-trivial curvature—is the *basal factuality* within the curvature-first ontology; reproduction (C3) and conditional debt (C4) then ensure that this foundation *continues* to obtain as a fact. This distinction is critical: CNT is not a "process ontology," but a "geometric ontology incorporating processual evolution and maintenance conditions." Existence is not the product of process; but the evolution and persistence of existence are the result of process.

2.2 The Tripartite Ontology of Spacetime

CNT adopts a specific ontology of spacetime, distinct from both substantivalist and relationalist traditions:

Property	**Description**	**Status**
Independence	Spacetime exists independently of closed nuclei	Background structure
Continuity	At macroscopic scales, spacetime is smooth	Classical limit
Flatness (ground state)	The ground state of spacetime is Minkowski	Vacuum definition
Curvature	Curvature is intrinsic; deviation from flatness is excitation	Ontologically prior

This is a *background-dependent* framework, in the sense that a flat Minkowski background is posited as the ground state, yet it is also *background-generating*, in the sense that the curved spacetime of general relativity emerges from the collective structure of excitations. The relationship to general relativity is established in §2.4. Crucially, the above table reflects the curvature-first ontology: curvature is not *produced by* closed nuclei; rather, closed nuclei *are* local configurations of intrinsic curvature. The standard GR picture—curvature "sourced by" mass-energy—is inverted: mass-energy is the readout of how curvature binds energy quanta.

2.3 Mass as an Emergent Quantity

In CNT, mass is not a fundamental property of matter. It is an *emergent quantity*—the readout of a process driven by intrinsic curvature:

1. Energy quanta (each carrying energy $h\nu$) move along geodesics of the background spacetime. The geodesics are determined by intrinsic curvature—curvature is the *cause* of motion, not the consequence of mass.
2. When these quanta encounter the boundary of a closed nucleus, they are absorbed.
3. The absorbed energy is converted to equivalent mass via $E = mc^2$.
4. The mass of a closed nucleus is the cumulative readout of all absorbed energy quanta:

$$m = \frac{1}{c^2} \sum_i h\nu_i = \frac{h\nu}{c^2} \int_S \rho \, dA \tag{2}$$

where ρ is the information density on the closed nucleus boundary S.

This mechanism separates two concepts that are conflated in the standard model:

Concept	Standard Model	CNT
Mass	Intrinsic property of matter	Emergent readout of absorbed energy quanta
Inertia	Resistance to acceleration	Identical to mass (equivalence preserved)
Gravitational charge	Identical to inertial mass	Not a charge; gravity is a curvature response

The equivalence of inertial and gravitational mass is not a mystery in CNT—it is a consequence of the fact that both are readouts of the same underlying process of energy-quantum absorption. Inertial mass is the energy content of the closed nucleus; gravitational "charge" is spacetime curvature's response to the quantized stress-energy tensor of the same energy quanta.

2.4 Gravity as Ontological Response

In CNT, gravity is a force—but fundamentally different from the three gauge forces (electromagnetic, strong, weak). It is not mediated by gauge bosons. It is an *ontological response*—spacetime curvature responds directly to the presence of the quantized stress-energy tensor.

The stress-energy tensor of a closed nucleus is not a continuous classical field, but a *quantized* sum over the energy quanta bound within the nucleus:

$$T_{\mu\nu} = \sum_i h\nu_i \, u_\mu^{(i)} u_\nu^{(i)} \tag{3}$$

where $u_\mu^{(i)}$ is the four-velocity of the i-th energy quantum and $h\nu_i$ is its energy. Spacetime curvature responds to this quantized source through the Einstein field equations in their standard form:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{4}$$

The physical picture of this binding mechanism is as follows: closed nuclei are curvature excitations; energy quanta move along geodesics of the background spacetime—both respond to curvature directly and independently. The energy quanta are absorbed at the boundary of the closed nucleus, and the gravitational constant G is the *binding coupling constant* measuring how strongly energy quanta are bound within each closed nucleus. The classical stress-energy tensor $T_{\mu\nu} = \sum_i h\nu_i u_\mu^{(i)} u_\nu^{(i)}$ is the effective description of this binding; through the Einstein field equations, this $T_{\mu\nu}$ then aligns with the curvature of the closed nucleus that performs the binding. Different closed nuclei all share the same curvature field—closed nuclei are themselves curvature excitations, each contributes its own curvature to the total curvature field, and each simultaneously responds to the total curvature produced by all. This synchronous, curvature-mediated interaction among all closed nuclei is the ontological essence of gravity, which differs fundamentally from gauge forces mediated by discrete boson exchange. G is not fundamental but emerges from the geometric structure of the closed nucleus network. Deriving G from the underlying 4-simplex geometry is an open problem (§6.1).

This formulation achieves a clean separation of responsibility: general relativity remains the correct description of how spacetime curvature relates to the stress-energy tensor; CNT supplies the ontological origin of the stress-energy tensor itself. The left-hand side (geometry) and right-hand side (matter) of the Einstein equations are no longer independent—the right-hand side is a particular configuration of the fundamental excitations of the left-hand side.

2.5 Compatibility with Special Relativity

The compatibility of CNT with special relativity is not a supplementary argument but a necessary structural consequence of the closed nucleus ontology. To understand this compatibility, one must first recognize that *time* in CNT is not an external coordinate parameter t—it is the iteration of the reproduction morphism $\mu : S \to S$. The closed nucleus does not "move through" time; it *generates* time via the discrete sequence of reproduction events $\mu \circ \mu = \mu$.

2.5.1 Time Dimension: Iteration of Reproduction

In the closed nucleus ontology, the time dimension is constituted by the iteration of the reproduction morphism:

$$t_k = k \cdot T_{\text{rep}} \tag{5}$$

where T_{rep} is the period required to complete one reproduction operation μ. The non-invertibility of the conditional debt operator ε (condition C4, $\neg\text{IsIso}(\varepsilon)$) guarantees strict monotonic increase, endowing the sequence with an arrow of time. Time is therefore not a background parameter but the *intrinsic rhythm* of the reproduction morphism's iteration.

2.5.2 Emergence of the Speed of Light

In the isolated closed nucleus stage, the concept of an "information propagation speed" has no operational meaning—each closed nucleus independently performs reproduction, and its products carry formal markers without network connectivity.

When multiple closed nuclei become connected through their reproduction products, the network imposes geometric constraints. Information propagation is confined within the 4-simplex structure of each closed nucleus. The maximal propagation distance is the 4-simplex diameter $D = \sqrt{2}\,\ell_0$ (guaranteed finite by compactness), and the minimal propagation time is one reproduction period T_{rep}. The speed of light emerges as the upper bound:

$$c = \frac{D}{T_{\text{rep}}} = \sqrt{2}\,\ell_0 \cdot f_{\text{rep}}, \quad f_{\text{rep}} = \frac{1}{T_{\text{rep}}} \tag{6}$$

Thus c is not a pre-network fundamental constant, but an emergent quantity arising from the geometric structure (ℓ_0) and dynamic rhythm (T_{rep}) of the closed nucleus network. The compactness of the 4-simplex (finite diameter) is the geometric basis for c being bounded. This relation unifies spatial geometry, temporal dynamics, and the information propagation upper bound into a single coherent emergent structure.

2.5.3 Three Compatibility Mechanisms

CNT's compatibility with special relativity rests on three mechanisms rooted in the closed nucleus's intrinsic structure:

(1) **Lorentz contraction → curvature enhancement (spatial mechanism).** When a closed nucleus moves at speed v relative to an observer, Lorentz contraction compresses its spatial scale along the direction of motion: $\ell(v) = \ell_0/\gamma$, where $\gamma = 1/\sqrt{1 - v^2/c^2}$. The 4-simplex dihedral angle $\Theta = \arccos(1/4)$ remains unchanged, hence the local curvature radius contracts as $\mathcal{R}(v) \sim \mathcal{R}_0/\gamma$, and the Gaussian curvature increases as $K(v) \propto \gamma^2 K_0$. This curvature enhancement is not a coordinate illusion but physically real: the closed nucleus must maintain the same reproduction operation ($\mu \circ \mu = \mu$) in a contracted volume, forcing its intrinsic geometry to become more tightly curved. The temporal curvature concomitantly adjusts as $g'_{00} = -\gamma^2$, which directly modulates the reproduction period (see mechanism 3).

(2) **One nucleus, one horizon: rigidity of spatial capacity.** Each closed nucleus possesses a fixed number of visible faces (3 out of 5 tetrahedral faces of the 4-simplex), with total area $A_{\text{visible}} = (3\sqrt{3}/4)\ell_0^2$. This area is a topological invariant—it does not change under Lorentz transformations. Combined with the universal information density limit $\rho_{\max} = 1/(4\ell_P^2 \ln 2)$ set by the Planck length ℓ_P, every closed nucleus has the same maximal information capacity $N_{\max} = \rho_{\max} \cdot A_{\text{visible}}$. Rest mass, given by $m_0 = (hf_{\text{rep}}/c^2)N_{\max}$, is strictly invariant under motion because spatial capacity does not change. Mass differences between nuclei arise only from the filling factor η (actual occupied information), not from geometric capacity variation. This guarantees the Lorentz invariance of rest mass as required by special relativity.

(3) **Time dilation = real change in reproduction period.** The temporal curvature modulation (mechanism 1) directly lengthens the reproduction period:

$$T_{\text{rep}}(v) = \gamma\, T_{\text{rep}}(0) \tag{7}$$

This is not a coordinate artifact but a physical change in the period required to complete $\mu \circ \mu = \mu$. All physical processes of the closed nucleus—particle decays, quantum oscillations, information processing, biological metabolism—are accumulations of reproduction events. With T_{rep} genuinely extended, fewer reproduction events occur per unit coordinate time. The twin paradox receives a microscopic resolution: the traveling twin's closed nuclei undergo fewer reproduction events ($N_v = N_0/\gamma$), hence experience fewer physical events; the traveling twin is genuinely younger. The traveler does not feel slowed because all internal processes—clocks, atoms, metabolism, consciousness—run at the same reproduction period γT_0. The number of reproduction events defining a "felt second" is the same on Earth and in the spaceship; only the global comparison reveals the difference.

The unification with gravitational time dilation is immediate: gravitational curvature modulates the temporal metric component g_{00} through the Einstein field equations, producing $T_{\text{rep}}(r) = T_\infty/\sqrt{1 - 2GM/r}$. Both special-relativistic and gravitational time dilation are thus the same geometric mechanism—four-dimensional spacetime curvature modulating the reproduction period—realized in different contexts. The equivalence principle is a natural consequence of this unified geometric dynamics. The rigorous identification of T_{rep} as the fundamental proper time parameter is developed in §3.2.1.

2.6 Ontological Status of the 4-Simplex

The regular 4-simplex is the minimal geometric realization of a closed nucleus. Its ontological status requires careful specification:

- The 4-simplex is *not* a fundamental "building block" in the atomistic sense. It is a *geometric idealization*—the simplest spacetime configuration satisfying the five criteria above.
- Natural (non-regular) 4-simplices correspond to excited states of closed nuclei. The regular case is the ground state.
- The 4-simplex geometry is fundamentally *discrete* at the base level. The continuous spacetime of general relativity emerges as a coarse-grained limit, analogous to how fluid dynamics emerges from molecular dynamics.

The geometric properties of the regular 4-simplex relevant to the CNT framework are developed in §4.1.

3 Methodological Framework

3.1 Dynamic vs. Static Analysis

CNT adopts a distinctive methodological framework that distinguishes two modes of analysis:

Mode	Object of study	Method
Dynamic analysis (structural tension $\to$ quantity-quality $\to$ hierarchical emergence)	Structure formation, emergence, history	Genetic/logical derivation
Static analysis (cyclology + tomography)	Stable structures, conservation laws	Mathematical formalization

Dynamic analysis asks: how does a physical structure come into being? It traces the genetic chain from the simplest possible configuration to the complex structure under investigation. This mode corresponds to the three links of the dialectics of nature—structural tension drives evolution, quantity-to-quality transition (p-adic stratification) marks evolutionary stages, hierarchical emergence (RG fixed points) produces irreducible new structures—constituting a rigorous historical-genetic logic.

Static analysis asks: given a stable structure, what are its invariant properties? It employs mathematical tools such as category theory, algebraic topology, and p-adic analysis to formalize the conserved quantities of that structure.

The two modes are complementary. Dynamic analysis provides the *why*—the historical reason a structure has the form it does. Static analysis provides the *what*—the precise mathematical characterization of that form. The framework requires both, and neither is reducible to the other.

3.2 Reproduction: The Elevation from Weak to Strong Ontology

Reproduction is not an independent axiom but a structural consequence of taking curvature-first ontology seriously. The logic is as follows: if curvature is intrinsic and mass is the readout of bound energy quanta, then the continued existence of mass requires that the geometric conditions binding energy quanta be *maintained*. Binding is not a one-time event—energy quanta move along geodesics, and the boundaries that trap them must be continuously regenerated. This regeneration of binding conditions is *reproduction* in the precise sense. Its core mechanism is:

- **Producing again**: conditions are *produced again*, not continuously generated. The operation is discrete, iterative.
- **Conditions of maintenance**: what is produced is not the structure itself, but the *conditions* that enable the structure to persist.
- **No additional requirements**: the paradigm does not require that successive conditions differ, that proliferation occurs, that history be preserved, or that irreducibility holds. These are *protective belt* options that a specific construction may choose or reject.

The mathematical expression of reproduction is the idempotent morphism:

$$\mu : S \to S, \quad \mu \circ \mu = \mu \tag{8}$$

The idempotence condition $\mu \circ \mu = \mu$ has a precise physical meaning: *repeating the reproduction operation does not alter the formal identity of the structure.* This distinguishes reproduction from *replication* (which would produce a different structure) and from *growth* (which would alter the form of the structure).

The reproduction operator μ is non-invertible: $\neg\mathrm{IsIso}(\mu)$. This irreversibility is the mathematical origin of the thermodynamic arrow of time—each reproduction event irreversibly transforms the state, even though the formal structure remains idempotent.

3.2.1 Proper Time and the Reproduction Cycle

The reproduction operation μ is not merely an abstract algebraic morphism—it has a concrete temporal realization. Each application of μ corresponds to one reproduction event, and these events occur at discrete intervals of *proper time* along the closed nucleus worldline. The time dimension of a closed nucleus is the iteration of reproduction, $t_k = k \cdot T_{\mathrm{rep}}$, where T_{rep} is the fundamental reproduction period. The conditional debt operator ε (condition C4) guarantees the irreversibility of this sequence, endowing it with an arrow of time.

After the closed nucleus network forms (§2.5), the reproduction period T_{rep} and the 4-simplex diameter $D = \sqrt{2}\,\ell_0$ jointly determine the speed of light as $c = D/T_{\mathrm{rep}}$, and reciprocally, $T_{\mathrm{rep}} = D/c$. This establishes T_{rep} as the intrinsic proper time standard of the closed nucleus network—the time unit against which all physical evolution is measured.

Because T_{rep} is the intrinsic rhythm of the reproduction morphism, it is *Lorentz-invariant* in the sense that all inertial observers agree on the number of reproduction events a given closed nucleus has undergone. However, under relative motion, the reproduction period is physically modulated by Lorentz contraction. As established in mechanism 1 of §2.5, motion compresses spatial scale ($\ell(v) = \ell_0/\gamma$) and enhances intrinsic curvature ($K(v) \propto \gamma^2 K_0$), which in turn lengthens the reproduction period:

$$T_{\mathrm{rep}}(v) = \gamma\, T_{\mathrm{rep}}(0), \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{9}$$

This is the proper time expression in closed nucleus ontology: the reproduction period, not the Minkowski arc length, is the fundamental proper time parameter. In the vacuum (curvature-free) limit, T_{rep} reduces to the standard special-relativistic proper time interval.

The structural consequences of this identification are as follows:

- **Time parameter unification.** T_{rep} serves simultaneously as (a) the period of the reproduction cycle, (b) the minimal time unit for information propagation in the network ($c = D/T_{\mathrm{rep}}$), and (c) the intrinsic clock of the closed nucleus. The framework thus possesses a single, dynamically grounded temporal parameter rather than an externally imposed "physical time."

- **Time dilation as reproduction-period extension.** Mechanism 3 of §2.5: relativistic time dilation of a moving closed nucleus is a *real physical change* in T_{rep}, driven by curvature enhancement from Lorentz contraction. All processes—particle decays, quantum oscillations, biological metabolism—accumulate via reproduction events; fewer events per unit coordinate time mean less physical evolution. The traveling twin is genuinely younger because fewer reproduction events have occurred ($N_v = N_0/\gamma$). The traveler does not feel slowed: a "felt second" corresponds to a fixed number of reproduction events, and this number is identical on Earth and in the spaceship; only global comparison reveals the difference.

- **Unification with gravitational time dilation.** Gravitational curvature modulates g_{00} via the Einstein equations, producing

$$T_{\mathrm{rep}}(r) = \frac{T_\infty}{\sqrt{1 - 2GM/r}}$$

  Special-relativistic and gravitational time dilation are therefore the same mechanism—spacetime curvature modulating T_{rep}—in different geometric contexts. The equivalence principle follows naturally.

- **One nucleus, one horizon.** Each closed nucleus carries a fixed spatial horizon area A_{visible} that is topologically invariant under Lorentz transformations. Combined with the universal Planck-length information density limit, this guarantees that rest mass is strictly Lorentz-invariant (m_0 unchanged by motion), consistent with special relativity (mechanism 2 of §2.5).

The reproduction cycle thus completes the kinematic and dynamic foundations of the framework: special relativity is not an external requirement to be satisfied but a structural consequence of the closed nucleus's intrinsic geometry and its reproduction dynamics.

3.3 The Strong Construction: CNT's Protective Belt

CNT is a *strong construction* of Reproduction Physics. It adds four protective-belt hypotheses on top of the minimal paradigm:

(H1) **Proliferation**: the reproduction process can produce new closed nuclei, not merely sustain existing ones. This is the origin of particle multiplicity.

(H2) **Fission**: the proliferation of closed nuclei is not a continuous process but occurs at discrete, *prime-numbered* reproduction events. This is the origin of irreducibility.

(H3) **Irreducibility**: a fission event cannot be decomposed into sub-processes. This is the physical counterpart of the mathematical fact that prime numbers are multiplicatively indecomposable.

(H4) **Historical dependence**: the state of a closed nucleus depends on its complete reproductive history. Higher-level structures are not reducible to a simple superposition of lower-level information—this is the irreducibility of history.

These hypotheses are *not* part of the hard core of the Reproduction paradigm (here the "hard core" is the curvature-first ontological inversion and the reproduction mechanism deduced from it). They are CNT-specific choices that can be modified or rejected without violating the paradigm.

The relationship between Reproduction Physics and loop quantum gravity (LQG) deserves explicit statement, because it bears directly on the framework's detectability within the quantum gravity community. The covariant spin-foam formulation of LQG—particularly the EPRL vertex amplitude [19]—is a construction that naturally falls *within* the Reproduction Physics paradigm, for the following structural reasons:

(1) **Geometric ontology.** LQG treats spacetime geometry as the fundamental ontological entity, with states represented by spin networks (graphs labeled by $SU(2)$ representations) and evolution by spin foams (2-complexes whose faces carry $SU(2)$ quantum numbers). This geometric ontology aligns with the curvature-first ontology of Reproduction Physics: curvature is not caused by matter but is intrinsic to spacetime.

(2) **Discrete geometric excitations.** The fundamental building block of a spin foam is the 4-simplex—the same geometric structure that CNT identifies as the minimal closed nucleus. In LQG, each 4-simplex in the triangulation carries a quantum of volume; in CNT, each 4-simplex is a closed nucleus—a local excitation of spacetime curvature. The structural identity of the fundamental geometric cell in both frameworks is not a coincidence but a manifestation of their shared geometric ontology.

(3) **Reproduction as evolution.** Spin-foam evolution maps a boundary spin network state to another through the insertion of internal vertices and faces—this is an iterative, discrete process. In the language of Reproduction Physics, each step of spin-foam evolution is a reproduction event: it produces again the conditions that sustain the geometric structure. The idempotence $\mu \circ \mu = \mu$ of the reproduction morphism corresponds to the stability of the spin-foam amplitude under coarse-graining (cylinder consistency).

(4) **Weak vs. strong construction.** Standard LQG spin-foam models satisfy the reproduction condition (iterative geometric evolution) and historical dependence (boundary states encode prior evolution), but do not include the CNT-specific strong-construction hypotheses: proliferation (controlled production of new geometric excitations at prime-numbered iterations), fission (irreducible prime decomposition of proliferation events), and irreducibility (non-decomposability of individual fission events). LQG is therefore a *weaker* construction within the same Reproduction Physics paradigm—equally legitimate but less structurally constrained than CNT. The two constructions are not competitors but complementary developments of the same paradigm.

This explicit identification is not a claim that LQG "belongs to" CNT, nor that LQG researchers have adopted Reproduction Physics. It is the structural observation that LQG's spin-foam formulation instantiates the minimal conditions of the Reproduction paradigm—curvature-first ontology, discrete geometric evolution, and iterative structure maintenance—and therefore constitutes an independent weak construction within the same paradigm. The identification serves the dual purpose of situating CNT within the landscape of existing quantum gravity research and inviting cross-fertilization between the two frameworks.

3.4 Principles of Mathematical Tool Selection

CNT's selection of mathematical tools follows the principle of *conceptual necessity* rather than convenience:

- A mathematical structure is admitted into the framework only if it emerges *necessarily* from the physical concepts, not because it happens to be computationally convenient.

- The chain of necessity must be traceable: each mathematical structure must follow from antecedent physical concepts through a logical step that admits no alternative.
- If a mathematical structure could be replaced by another without altering the physical content, it is not part of the framework's hard core—it is a representational choice.

This principle leads to the identification of p-adic numbers as the necessary mathematical direction of Reproduction Physics, as developed in §4.2.

3.5 The Concept Chain

The logical structure of CNT can be expressed as a concept chain, where each concept follows necessarily from the preceding one. The chain does not begin with reproduction as an arbitrary axiom, but with the ontological inversion derived from general relativity:

$$\boxed{\begin{array}{l}\text{Curvature-first ontology: } G_{\mu\nu} \text{ prior } \Rightarrow \text{ Geodesic binding of energy quanta} \\ \quad \Rightarrow \text{ Reproduction } \mu \circ \mu = \mu \Rightarrow \text{ Prime marking } p \in \mathbb{P} \\ \quad \Rightarrow p\text{-adic encoding } \mathbb{Z}_p \Rightarrow \text{ Holographic boundary emergence} \\ \quad \Rightarrow \text{ Gauge groups } \Rightarrow \text{ Role of charges } \Rightarrow \text{ Irreducible dependence}\end{array}} \tag{10}$$

The chain is not a postulate but a *derivation trace*. Each arrow represents a logical inference compelled by the physical content of the antecedent concept. The chain is falsifiable: if any inference can be shown to admit a physically distinct alternative, the chain breaks at that point. Crucially, the chain is falsifiable at its root: if the curvature-first ontology is wrong—if mass-energy is ontologically prior to curvature—the entire chain dissolves. There is no fallback position.

3.6 Cycle-Tomography Spacetime Duality

The methodological framework of CNT is completed by a fundamental duality: *cycle analysis (cyclology) is time-structure analysis; tomography is space-structure analysis.* "Tomography" is chosen deliberately—it means cutting and dissecting layer by layer, extracting the independent structure of each layer, as in tomographic imaging.

3.6.1 Two Complementary Analytical Dimensions

Dimension	Analyzes	Key questions
Tomography (space structure)	Range, force carriers, mechanical form	"What does this force look like?"
Cyclology (time structure)	Coupling strength, maintenance mechanism, frequency	"How does this force stay alive?"

Tomography answers: what is the force's range? What particles mediate it? What is its mathematical form? These correspond directly to the gauge group structure $SU(3) \times SU(2) \times U(1)$, the gauge bosons (gluons, W/Z, photon), and the representation structure. Tomography operates through the projective limit structure of p-adic numbers: $\mathbb{Z}_p =$

$\varprojlim \mathbb{Z}/p^n\mathbb{Z}$, where the projection maps $\pi_n : \mathbb{Z}/p^{n+1}\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$ perform the layer-by-layer dissection, extracting the independent structure of each hierarchical level.

Cyclology answers: what is the coupling strength? How is the force sustained? What is its reproductive frequency? The defining mechanism is the *iterative synthesis of forces through reproduction cycles*: a force is not a single event but a bundled synthesis of prime-numbered reproduction cycles. p consecutive reproduction events are bundled into one irreducible *action packet*—this is the quantitative origin of the strength of that force. The relationship is not simple, because multiple competing factors are at work—including but not limited to the number of events per packet, the packet density per unit time, and the algebraic-structural differences between primes—and the net coupling strength emerges from their mutual constraints. The precise functional form is precisely what the "computational hub" problem aims to determine. The synthesis mechanism maps to the coupling constants ($\alpha \approx 1/137$, $\alpha_s \approx 1$, $g_W \approx 0.65$), where each force is labeled by a distinct prime p. The idempotence of the reproduction morphism $\mu \circ \mu = \mu$ ensures that the bundled packet remains stable under iteration—the force does not decay but is maintained. Cyclology operates through the valuation structure of p-adic numbers: $\nu_p : \mathbb{Q}^\times \to \mathbb{Z}$, which counts reproduction events per cycle, encodes the depth of synthesis (prime power), and marks the cyclic history.

3.6.2 The Gauge Group as Intersection Point

The two analytical dimensions are not orthogonal—they intersect at the *gauge group*. This is the central structural insight of the duality:

- **Tomography gives the "skeleton" of the gauge group**: which groups exist ($SU(3)$, $SU(2)$, $U(1)$), their representation structure (color, weak isospin, hypercharge), and the force carriers (gluons, W/Z, photon).
- **Cyclology gives the "blood" of the gauge group**: the historical order of group emergence ($p = 2 \to p = 3 \to p = 5 \to \cdots$), the traversal mechanism by which each later group interacts with all earlier groups, and the coupling strengths determined by cycle depth—essentially how many reproduction events are bundled per action packet.

The gauge group is therefore not merely a kinematical input—it is where the two fundamental analytical dimensions converge. This convergence is the reason that the "computational hub" problem—finding the quantitative relation between historical cycle frequency and gauge interaction strengths—cannot be split into two independent subproblems. The real difficulty is not solving each dimension separately, but finding how they interlock at the gauge group.

3.6.3 Dual Role of p-Adic Numbers

The same p-adic number system plays two distinct roles in the two modes of analysis, testifying to its status as the natural language of Reproduction Physics:

Analysis mode	**Role of $\mathbb{Z}_p$**	**Mathematical operation**
Tomography (space)	Encoder of hierarchical structure	Projective limit $\varprojlim$
Cyclology (time)	Counter of reproductive history	Valuation map ν_p

This dual role is not a coincidence but a manifestation of the deep unity of Reproduction Physics: the spatial encoding of hierarchical structure and the temporal counting of reproductive events are two aspects of the same p-adic mathematical object.

3.6.4 Relation to Standard Quantum Field Theory

Standard QFT is predominantly a *tomographic* (spatial-structure) theory. Its core achievements include the gauge group structure, the gauge boson content, the Lagrangian form, and Feynman diagram topology. All of these belong to tomography: they describe *what* the force looks like. QFT's blind spot is precisely the cyclological dimension: it cannot explain *why* the gauge group is $SU(3) \times SU(2) \times U(1)$, *why* the coupling constants take their observed values, or what the historical order of gauge group emergence is. CNT is not a replacement for QFT but its completion: QFT supplies the mature tomographic form; CNT adds the missing cyclological dimension and unifies both at the gauge group.

3.6.5 Position in the Dialectics of Nature

Cycle-tomography duality is not a fourth link independent of the three links of the dialectics of nature (structural tension, quantity-quality transition, hierarchical emergence). Rather, it is the *concrete instantiation* of the historical-hierarchical analysis method that runs through all three links. Tomography (space-structure analysis) and cyclology (time-structure analysis) jointly constitute the operative analytical machinery at each dialectical stage: structural tension manifests as the irreconcilable geometric constraint driving reproduction, quantity-quality transition is tracked by p-adic valuation shifts, and hierarchical emergence is described by the projective limit structure of tomographic layers.

3.6.6 Detailed Layered Decomposition

The two analytical dimensions each decompose into three specific physical questions, with precise mappings to CNT structures:

Tomography (Space-Structure Analysis) "What does the force look like?"		
Dimension	**Physical content**	**CNT correspondence**
Range	Interaction distance	Hierarchical depth of gauge groups
Force carriers	Particles mediating interaction	Gauge bosons (photon, gluons, W/Z)
Mechanical form	Mathematical expression	Representation structure ($SU(3)$, $SU(2)$, $U(1)$)

Cyclology (Time-Structure Analysis) "How does the force stay alive?"		
Dimension	**Physical content**	**CNT correspondence**
Coupling strength	Magnitude of interaction	$\alpha \approx 1/137$, α_s, g_W emerge from historical cycle frequency via competing action-packet synthesis effects
Maintenance	How the force persists	Idempotence $\mu \circ \mu = \mu$
Cycle frequency	Density of cycles	Historical cycle frequency f_{cycle}

The key structural insight from this layered decomposition is operational, not merely taxonomic. Tomography dissects *what is present* in the spatial hierarchy—the "anatomy" of the force; cyclology tracks *what persists* through temporal iteration—the "physiology" of the force. The gauge group is the only entity that answers both questions simultaneously, which is why it serves as the convergence point between the two dimensions.

3.6.7 Decomposition of the "Computational Hub" Problem

The core missing piece of CNT is the "computational hub"—the quantitative relation between historical cycle frequency and gauge interaction strengths:

$$f_{\text{cycle}} = f(g_s, g_w, e) \tag{11}$$

Based on the cycle-tomography duality, this problem decomposes into two sub-problems that are methodologically distinguishable but not independent:

Problem	**Analytical tool**	**Mathematical language**	**Objective**
Spatial problem	Tomography	Projective limit $\varprojlim$	Derive gauge group structure
Temporal problem	Cyclology	Valuation map ν_p	Derive coupling constants

- **Spatial problem (tomography):** Why is the gauge group $SU(3) \times SU(2) \times U(1)$? Derived from the fission logic of the tomographic structure.
- **Temporal problem (cyclology):** Why do the coupling constants take their observed values? ($\alpha \approx 1/137$, $\alpha_s \approx 1$, $g_W \approx 0.65$). Derived from the cycle-frequency logic.

The non-independence of these two sub-problems is critical. The gauge group structure (spatial problem) determines the traversal pattern through which each later-emerging group interacts with all earlier groups (temporal problem), and the traversal pattern in turn constrains the allowable gauge group structure. The real challenge of the computational hub is not solving each sub-problem separately, but finding how they *interlock* at the gauge group—making the convergence path (§3.6.8) the decisive one.

3.6.8 Methodological Significance: Three Breakthrough Paths

The cycle-tomography duality illuminates three distinct paths toward theoretical breakthrough:

- **Path 1—Tomographic breakthrough:** Derive the gauge group structure $SU(3) \times SU(2) \times U(1)$ from the fission logic of reproduction. The key question is how prime-numbered fission produces precisely these three groups in direct product.
- **Path 2—Cyclological breakthrough:** Derive the coupling constants α, α_s, g_W from the cycle frequency f_{cycle}. The key question is how the density of reproductive cycles determines the strength of gauge interactions.
- **Path 3—Convergence breakthrough** (the most critical): Realize the unification of tomography and cyclology at the gauge group. The key question is how the historical order of gauge group emergence simultaneously determines both structure and coupling strength.

This three-path structure mirrors Einstein's situation in a revealing way. Einstein knew gravity was geometry (concept) but needed to find the field equation's form (mathematics) by exploring equations within existing Riemannian geometry. CNT knows that reproduction in an independent spacetime background generates gauge structure (concept), but must derive the specific form of that structure (mathematics) by "growing" it from the reproduction process itself. The cycle-tomography duality provides the direction of this growth: spatial structure grows from tomographic analysis, temporal structure from cyclological analysis, and their unification grows at the gauge group.

Standard QFT has already developed the tomographic dimension to maturity; CNT's task is to develop the cyclological dimension and achieve convergence at the gauge group. CNT's success therefore does not depend on finding a single "master equation"—it depends on making progress along each of the three paths, with the convergence path being the decisive one.

4 Mathematical Structure

4.1 Regular 4-Simplex Geometry

The regular 4-simplex is the minimal geometric realization of a closed nucleus. Its properties are entirely determined by the S_5 symmetry group of its five vertices.

The dihedral angle $\Theta = \arccos(1/4)$ is a *purely geometric fact*—it follows from the S_5 symmetry of the regular 4-simplex without any physical input. This angle appears repeatedly in the framework as a geometric anchor for physical quantities.

The 4-simplex is not an arbitrary choice. It is the minimal simplex capable of enclosing a four-dimensional region, and its S_5 symmetry group is the smallest non-abelian symmetry capable of supporting the full structure of the Standard Model gauge groups. The restriction to four spacetime dimensions is thus not an input but a consequence of the minimality requirement.

The regular 4-simplex is embedded in the flat Minkowski background spacetime as the ground state (§2.4), and consequently its metric signature is *Lorentzian* ($(-,+,+,+)$ or equivalently $(+,-,-,-)$). This signature is not an independent hypothesis—it is

Table 1: Geometric properties of the regular 4-simplex. ℓ_0 is the fundamental length scale.

Quantity	**Symbol**	**Value**
Edge length	ℓ_e	$\sqrt{2}\,\ell_0$
Dihedral angle	Θ	$\arccos(1/4) \approx 1.318$ rad
Number of vertices	–	5
Number of edges	–	10
Number of triangular faces	–	10
Number of tetrahedral cells	–	5
Symmetry group	–	S_5
Total surface area	A_{total}	$5\sqrt{3}\,\ell_0^2/4$

a consequence of the structural requirement that non-trivial excitations (closed nuclei) can only be defined in a Lorentzian background: a Euclidean signature does not admit timelike geodesics, making it impossible to support the geodesic motion of energy quanta (Equation E1) and causal reproduction dynamics. In other words, a closed nucleus is a spacetime *excitation* precisely because it deviates from the flat Minkowski ground state; its intrinsic geometry necessarily inherits the Lorentzian signature structure of that ground state.

A conjectural consequence of the 4-simplex geometry concerns the fine-structure constant. Using the EPRL spin-foam model [19] with face spin $j_f = 1/2$ and the correspondence $e_0 = \sin(5\Theta)$ linking the bare electromagnetic coupling to the total dihedral phase 5Θ accumulated over the 10 triangular faces, one obtains:

$$\boxed{\alpha_0 = \frac{\sin^2(5\Theta)}{4\pi} = \frac{375}{16384\pi}}, \qquad \boxed{\frac{1}{\alpha_0} = \frac{16384\pi}{375} \approx 137.258} \tag{12}$$

The experimental value is $1/\alpha = 137.035999084$ (PDG 2024 [20]), a deviation of $+0.162\%$. The exact algebraic expression follows from $\cos\Theta = 1/4$ and the Chebyshev identity $\cos(5\Theta) = 61/64$, yielding $\sin^2(5\Theta) = 375/4096$.

4.2 p-Adic Numbers as the Necessary Mathematics of Reproduction Physics

A valuable theoretical system should grow its needed mathematics from within—this is a foundational methodological principle of CNT (§3.3). The identification of p-adic numbers as the necessary mathematical direction of Reproduction Physics is one of the core insights of the CNT framework [14]. It is not an arbitrary choice but a *derivation*:

1. **Reproduction ⇒ natural numbers.** Reproduction is an iterative operation. Counting reproduction events generates the natural numbers $\mathbb{N}$: the 1st reproduction, 2nd reproduction, 3rd reproduction, ...

2. **Fission ⇒ prime numbers.** In CNT's strong construction, prime-numbered iterations of reproduction events trigger fission—the production of new closed nuclei. Prime numbers are the multiplicative atoms of $\mathbb{N}$: they cannot be decomposed into smaller reproduction events. Hence, fission at prime-numbered iterations is *irreducible*—it cannot be decomposed into sub-processes.

3. **Irreducibility $\Rightarrow$ p-adic valuation.** An irreducible fission event at a prime p can be assigned a p-adic valuation $v_p(x)$ that measures "the number of times p divides the reproduction count." This valuation encodes the hierarchical depth of fission.

4. **Historical dependence $\Rightarrow$ historical encoding.** In CNT's strong construction, higher reproduction levels retain information about lower levels. The mathematical expression of this information has two candidate forms, corresponding to two p-adic structures: **standard p-adic numbers** encode history via the compatible-sequence projective limit $\mathbb{Z}_p = \varprojlim \mathbb{Z}/p^n\mathbb{Z}$—lower-level information is folded into the compatibility conditions of the limit construction; **compositional p-adic numbers** encode history via the forward generative construction $x = \sum S_n P_n$—the complete reproduction record of every level is unfolded into the overall structure, with lower-level history folded into the cumulative products $P_n = \prod p_i$. CNT's strong construction selects the compositional p-adic forward encoding, motivated by its injectivity/multi-valued asymmetry: in compositional p-adic numbers, the map from coefficient sequence to x is injective (sequence $\mapsto x$ is unique), but the decoding from x back to the coefficient sequence is multi-valued (x cannot uniquely recover the coefficient sequence)—this is the mathematical expression of the *irreducibility of history*: a single observable x can correspond to multiple distinct reproductive histories; history is more fundamental than the numerical value. Note: in compositional p-adic numbers, $x \bmod P_k$ cannot uniquely recover lower-level coefficient sequences due to the unbounded nature of $a_{n,i} \in \mathbb{N}$—not information loss, but non-unique encoding format. The two p-adic structures share the same logical core—history cannot be forgotten—but differ in their encoding mechanism: standard p-adic numbers enfold history via limits, compositional p-adic numbers unfold history via generation.

This chain is logically necessary. Given the physical concepts of reproduction, fission, irreducibility, and historical dependence, the p-adic numbers are the *unique* mathematical structure that faithfully encodes all four. The real numbers $\mathbb{R}$—the standard number system of physics—encode none of them.

However, what the above derivation establishes is the *structural type* of p-adic numbers (level fission, irreducibility, historical encoding, reproduction counting) as the necessary mathematical direction of Reproduction Physics, not their specific form. Standard p-adic numbers (fixed prime base, bounded coefficients) satisfy the logical chain, but whether they fully capture all the features of Reproduction Physics—in particular the endogenous generation of bases ($p_{n+1} = S_n$) and the non-forgetability of history—is a question requiring further exploration.

4.3 Compositional p-Adic Numbers: An Exploratory Construction

A known limitation of standard p-adic numbers is the externally given nature of the base p, which stands in tension with the requirement of Reproduction Physics that the base be generated from the sum of prior-level reproductions. Compositional p-adic numbers are a candidate form currently under exploration, whose core innovations are:

1. **Variable base**: $p_{n+1} = S_n$, the base is generated endogenously from the sum of prior-level reproductions;

2. **Natural number coefficients**: $a_{n,i} \in \mathbb{N}$, recording the count of each reproduction event with no upper bound;

3. **Level fission**: level transitions are $S_n = \sum a_{n,i} p_n^i$, a qualitative structural change from a layer to the next higher base; the historical encoding is the forward construction $x = \sum S_n P_n$, not the projective limit $\varprojlim \mathbb{Z}/P_n\mathbb{Z}$ (which, though formally well-defined since $P_n \mid P_{n+1}$, is information-losing due to the non-uniqueness of decoding).

A foundational feature is that **layer 0 is the natural numbers $\mathbb{N}$ themselves**—pure reproduction counts without any base; the first base $p_1 = S_0$ (plain sum of layer-0 reproduction counts) marks the first emergence of p-adic structure, which grows from layer 1 onward.

Compositional p-adic numbers satisfy three core properties: higher levels depend on all lower levels, deterministic historical encoding, and **bidirectional remainder**. In compositional p-adic numbers, the mathematical sign of remainder is a non-zero result of modular operation, with two types:

- **Structural remainder**: $x \bmod P_k = S_0 \neq 0$, the higher level (using cumulative product P_k) cannot exhaust the lower-level history;

- **Coefficient remainder**: $S_n \bmod p_n = a_{n,0} \neq 0$, the finite base cannot fully incorporate the higher-level coefficient.

Remainder is not directional (not merely from lower to higher); it is structural—the irreducible asymmetry of any part-whole relationship. This structure provides a mathematical candidate more closely aligned with CNT's ontological requirements.

A further structural feature is that the cyclic nature of reproduction (period-frequency duality) endows compositional p-adic numbers with a natural frequency-period dimensionality: via appropriate choice of temporal scale, $\nu \cdot T_0 = \sum_n (\nu_n T_0) \cdot P_n$ yields a dimensionless pure number, with the scale choice rooted in the cyclic ontology of reproduction rather than imposed externally. Their rigorous topological and analytic foundations are still under construction.

The remainder structure suggests two **conjectures**. First (**Conjecture: electron as remainder structure**): the electron may not be a closed nucleus, but a remainder structure: the fact that all electrons share identical mass and charge suggests they originate from the same level's coefficient remainder $a_{n,0}$, rather than from independent closed-nucleus constructions; the electron-to-proton mass ratio ($\sim 1/1836$) is naturally explained as a "debris" (the indigestible part of the primary structure's reproduction process) relative to the "body" (the closed-nucleus network). Second (**Conjecture: charge as remainder more fundamental than coupling constant**): the coupling constant belongs to the cycle-frequency–numerical-intensity analysis (a tomography quantity: the numerical ratio of different levels at the same instant), while charge belongs to the dynamical analysis of reproductive history (a cyclology quantity: the indigestible historical accumulation in the reproduction process), the two corresponding to the two sides of the cycle-tomography duality. Compositional p-adic numbers are an *exploratory construction*—they are not an established component of the CNT framework, but an active response to the open question of "the specific form of p-adic numbers." Their ultimate status depends on further mathematical development.

4.4 p-Adic / Real Duality

The number system of Reproduction Physics is not p-adic *instead of* real; it is p-adic *in addition to* real. Both $\mathbb{R}$ and $\mathbb{Q}_p$ are completions of the rationals $\mathbb{Q}$, distinguished by their valuations:

Property	**Real numbers $\mathbb{R}$**	**p-adic numbers $\mathbb{Q}_p$**
Completion of	$\mathbb{Q}$ under $\lvert\cdot\rvert_\infty$	$\mathbb{Q}$ under $\lvert\cdot\rvert_p$
Metric property	Archimedean	Ultrametric
Physical role	Continuous spacetime coordinates	Reproductive history encoding
Topology	Connected	Totally disconnected
Hierarchy	Continuous variation	Discrete tree structure

The ultrametric property of p-adic numbers—the strong triangle inequality $|x+y|_p \leq \max(|x|_p, |y|_p)$—is particularly significant. It implies a strict hierarchical (tree) structure without overlapping levels, precisely the structure needed to encode the stratified history of reproduction events.

4.5 Gauge Groups as Structural Layers

In CNT, the gauge groups of the Standard Model—$SU(3)_C \times SU(2)_L \times U(1)_Y$ [10–12]—are not fundamental inputs but distinct *structural layers* of the same reproduction structure. The three gauge forces—electromagnetism, the strong force, the weak force—are not independently existing entities. They are manifestations of different levels of the same reproductive dynamics. Their origin must be traced historically—through the evolutionary process by which these structural layers emerged—but this historical tracing is an analytical method, not an ontological claim about their present nature. This is a conjecture, not a confirmed result, but its logical structure is as follows:

1. The classification theorem for p-compact groups [18] establishes a one-to-one correspondence between primes p and admissible gauge group structures. Each prime labels a distinct possible gauge group.

2. In CNT, the primes appearing in the fission of reproduction events—for instance $p = 3$ (triple fission) and $p = 2$ (double fission)—suggest a natural correspondence with groups such as $SU(3)$ and $SU(2)$ respectively. This is an illustrative example of how the correspondence could work, not a unique assignment: the actual mapping from fission primes to gauge groups depends on the detailed fission structure, which remains undetermined ([**Gap 5**]).

3. The gauge group structure of the Standard Model then becomes a *projection* of reproductive history—the specific primes at which fission events occurred are recorded in the gauge group structure of the present universe.

This hypothesis is labeled [**Gap 5**] in §6. The correspondence between fission primes and gauge groups is a structural analogy, not yet a rigorous derivation. Its methodological significance, however, is clear: the apparent independence of gauge forces is a surface phenomenon—their unity is rooted in the identity of the reproduction structure that manifests as their distinct structural layers. This stance is naturally compatible with the requirement of grand unification suggested by the Standard Model. The Standard Model points toward the unification of its three gauge groups—$SU(3)_C$, $SU(2)_L$, and $U(1)_Y$—into a single simple group (e.g., $SU(5)$, $SO(10)$) at a high energy scale. In CNT's layered-structure picture, this "unification" is not a phenomenological coincidence at the high-energy limit but a consequence of their common structural origin: the three gauge groups are different structural layers of the same reproduction structure, and their apparent separation at low energies reflects the hierarchical depth of these layers. The driving force of grand unification is therefore not the running convergence of field-theoretic coupling constants, but the layered structure of the reproduction dynamics—high-energy unification is the physical manifestation of their shared structural root. This methodological stance is the philosophical foundation of the cycle-tomography duality developed in §3.6.

4.6 Reproduction and the Renormalization Group

An independent piece of evidence for the p-adic structure of Reproduction Physics comes from the renormalization group (RG). This connection is structural, not analogical:

- Wilsonian RG operates through coarse-graining—integrating out high-energy degrees of freedom layer by layer. This is a discrete, hierarchical operation, not a continuous one.
- The p-adic ultrametric property $|x + y|_p \leq \max(|x|_p, |y|_p)$ precisely describes this hierarchical structure: levels are strictly separated, with no overlap.
- In the p-adic $O(N)$ model [13], RG transformations are not differential equations but discrete steps between p-adic shells. RG fixed points are structurally identical to the reproduction idempotent operator $\mu \circ \mu = \mu$: both describe states invariant under the operation.
- This convergence of RG theory and Reproduction Physics on the same p-adic structure, from completely independent starting points, provides a non-trivial consistency check for the framework.

This connection remains an open problem for future investigation.

4.7 Historical Path Integral

The cycle-tomography duality (§3.6) finds its most concrete mathematical expression in the path integral. The standard QFT path integral $Z = \int \mathcal{D}\phi\, e^{iS[\phi]/\hbar}$ already contains both analytical dimensions in embryonic form: the action $S[\phi]$ encodes the mechanical form (tomography), while the integral over all field configurations $\int \mathcal{D}\phi$ sums over all possible histories (cyclology). However, the continuous integration measure $\int \mathcal{D}\phi = \lim_{N\to\infty} \int \prod_{i=1}^{N} d\phi(x_i)$ obliterates the discrete historical structure.

CNT's Historical Path Integral (HPI) is a *proposed construction* whose purpose is to restore this discrete historical structure. The conceptual requirements for the HPI are clear: (a) time steps must be non-uniform, reflecting the discrete reproduction events with step sizes $\Delta t_k = 1/f_k$ where f_k is the reproduction frequency at step k; (b) the integration measure must be the p-adic Haar measure rather than the Lebesgue measure, since the reproduction history space is naturally isomorphic to $\mathbb{Z}_p$ (§4.2); (c) in the limit $p \to \infty$, the HPI must reduce to the standard path integral of QFT. The rigorous construction of the HPI—including the p-adic path integral measure, the precise form of the p-adic corrected action, and the proof of the $p \to \infty$ limit—is a core open problem of the framework, designated [**Gap 11**]; it is cross-referenced here against §6 for contextual placement. Its completion would make explicit the discrete p-adic historical structure that the continuum limit of standard QFT obliterates, and would provide the mathematical foundation for the "computational hub" problem of §3.6: deriving coupling constants from reproductive history.

4.8 Working Hypothesis: Network Architecture of Closed Nuclei

The HPI (§4.7) provides the formal framework for a p-adic path integral, but leaves the *internal structure* of the historical evolution as a black box. The reproduction events, fission primes, and action-packet synthesis described in §3.6 operate *inside* this black box. To eliminate this opacity and open a concrete computational entry point into the p-adic path integral, a working hypothesis concerning the architecture of closed nuclei is proposed. This hypothesis is a **research tool, not a claim of truth**—it provides a concrete image for computation, but the actual historical process may differ fundamentally. The methodology is historical-structural analysis: reconstructing the evolutionary process from observable historical traces, which entails extensive trial and error.

4.8.1 From Single Closed Nucleus to Proton Network

The core assumption of the hypothesis is that the **proton is not a single closed nucleus but a network of closed nuclei**. The network is built from a large number of Planck-scale ($\ell_P = \sqrt{G\hbar/c^3}$) closed nuclei proliferating through reproduction. This image carries several structural advantages:

(1) **It demystifies historical evolution.** Historical evolution is not a metaphysical black box imposed from above, but the concrete outcome of actual reproductive practice—the proliferation, interaction, and stabilization of closed nuclei through successive reproduction cycles. The network architecture makes this process picturable and hence computationally accessible.

(2) **It provides an entry point for the p-adic path integral.** The proliferation from a single Planck-scale closed nucleus through multi-stage network formation is a discrete, countable process naturally encoded by p-adic numbers. Each stage of network growth corresponds to a level in the p-adic projective limit, and the reproductive counts at each stage determine the valuation structure.

(3) **It matches the quark sea image.** In QCD, the proton is not three static valence quarks but a dynamic "sea" of quark-antiquark pairs and gluons. A network of Planck-scale closed nuclei producing three distinguishable network components (the three quark types) as emergent organizational modes is structurally compatible with this image.

(4) **It provides compatibility with the existing definitions of G and c.** The relevant physical scale for the network's constituent closed nuclei is the Planck length

$\ell_P = \sqrt{G\hbar/c^3}$. That this length—constructed from G, c, and $\hbar$—appears naturally as the network's characteristic scale is not a reinterpretation of G or c, but a consistency check: the network architecture is compatible with these constants retaining their standard physical meanings (G as the energy quantum–closed nucleus binding coupling constant per §5.4, c as the emergent propagation speed from network geometry per §2.5). The Planck scale serves as the natural bridge between the known constants and the network's geometric structure.

4.8.2 Evolutionary Sequence with Structural Inheritance

The hypothesis proposes a concrete evolutionary sequence in which gauge structures emerge through *hierarchical inheritance*—each later stage self-consistently inherits the structures produced at all earlier stages. The essential principle is that there is only *one* real reproduction structure; the three gauge forces are its distinct structural layers. The sequence is as follows:

Stage 1 — Electromagnetic structure.
A single closed nucleus network (a "pre-proton" structure) establishes the electromagnetic gauge structure. This is the first-order structure: the reproduction dynamics at this stage mark $U(1)$ in the historical record.

Stage 2 — Weak structure (inheriting EM).
The network evolves into a two-component configuration (a two-quark network). The reproduction dynamics at this stage *self-consistently inherit* the first-order electromagnetic structure while generating the weak-force structure. The gauge record now contains both $U(1)$ and $SU(2)$.

Stage 3 — Strong structure (inheriting EM + weak).
The network further evolves into a three-component configuration (a three-quark network). The reproduction dynamics self-consistently inherit *both* the first-order electromagnetic structure and the second-order weak structure, while generating the strong-force structure. The gauge record now contains $U(1)$, $SU(2)$, and $SU(3)$.

Final stage — Proton/neutron as unified structure.
The three-component network stabilizes into the proton (or neutron, via isospin rearrangement). At this stage, the structure simultaneously carries: first-order EM ($U(1)$), second-order weak ($SU(2)$), third-order strong ($SU(3)$), and gravity—the last emerging from the collective curvature of the entire network, expressed at the proton scale through the stress-energy tensor $T_{\mu\nu}$ composited from the network's internal dynamics and aligning with the strong-force scale.

This sequence explains *why* gauge forces are structural layers rather than independent entities (§4.5): there is only one reproduction structure; each gauge group is a distinct structural layer of this same structure, coexisting in the present—their origin must be traced historically, but their present nature is layered, not historical.

4.8.3 Caveats and Methodological Status

This hypothesis is a computational tool, not a verified model. The actual historical process may differ in any or all of the following ways: the number of stages may be different, the order may be different, quark constituents may have formed *after* rather than before the proton, the network may not be organized by quark-type components at all, or the entire network image may be superseded by a different architecture.

The hypothesis is *not* proposed as truth but as a **methodological entry point**: it makes the "computational hub" problem of §3.6—deriving coupling constants from reproductive history—concrete enough to attempt computation. The guiding principle remains structural analysis informed by history: identify structural layers in observable gauge structures and trace their origin through historical reconstruction. This entails systematic trial and error across multiple candidate architectures. The network hypothesis is one such candidate, labeled [**Gap 13**] in §6.

5 Mathematical Requirements and Structural Constraints

5.1 The Six-Equation System

The mathematical skeleton of CNT is a system of mathematical self-consistency constraints that any physical realization of the Reproduction paradigm must satisfy. It traces the logical chain from the existence of energy quanta (E1) to the emergence of mass as a boundary readout (E5), with the Bekenstein bound (E6) serving as a universal saturation condition imposed by the holographic structure of reproductive history:

$$\boxed{\begin{array}{lll}
\text{(E1)} & \dfrac{d^2x^\mu}{d\tau^2} + \Gamma^{(0)\mu}_{\alpha\beta}\dfrac{dx^\alpha}{d\tau}\dfrac{dx^\beta}{d\tau} = 0 & \text{Energy quantum geodesic equation} \\
\text{(E2)} & \nabla^{(0)}_\mu J^\mu = 0 & \text{Energy quantum number conservation} \\
\text{(E3)} & \sigma(x,t) = J^\mu(x,t)\, n_\mu(x)\big|_{x\in S_t} & \text{Geodesic flux density} \\
\text{(E4)} & \dfrac{\partial\rho}{\partial t} = \sigma(x,t)\left(1 - \dfrac{\rho}{\rho_{\max}}\right) & \text{Information density evolution} \\
\text{(E5)} & m(t) = \dfrac{h\nu}{c^2}\displaystyle\int_S \rho(x,t)\sqrt{\gamma^{(0)}}\, d^2x & \text{Mass emergence} \\
\text{(E6)} & \rho_{\max} = \dfrac{1}{4\ell_P^2 \ln 2} & \text{Bekenstein bound}
\end{array}} \tag{13}$$

These equations form a strictly unidirectional causal chain:

$$\text{E1} \xrightarrow{\text{Defines } x_{\text{geo}}, u^\mu} \text{E3} \xrightarrow{\text{Defines } \sigma} \text{E4} \xrightarrow{\text{Defines } \rho} \text{E5} \xrightarrow{\text{Defines } m(t)} \tag{14}$$

Equation (E1) describes the free motion of energy quanta in the background spacetime. The background connection $\Gamma^{(0)}$ is the Christoffel symbol of the macroscopic metric $g^{(0)}_{\mu\nu}$, which is itself determined by the collective stress-energy tensor of all closed nuclei via the Einstein field equations. This forms a closed feedback loop: geometry determines geodesics, geodesics determine flux, flux determines mass, mass determines stress-energy, stress-energy determines geometry.

Equation (E2) enforces energy quantum number conservation. The current J^μ is constructed from the relativistic kinetic theory formalism [15–17], ensuring covariance and dimensional consistency.

Equation (E3) is the bridge between microscopic kinematics and macroscopic field theory. The flux density $\sigma(x,t)$ (dimensions $[L^{-2}T^{-1}]$) measures the rate at which energy quanta cross the closed nucleus boundary. The construction $J^\mu n_\mu$ ensures that only the normal component of the flux contributes.

Equation (E4) is the core dynamical hypothesis of the framework. The information density $\rho(x,t)$ is defined on the closed nucleus boundary S (a two-dimensional spatial surface), with t a parameter along the boundary worldline. This equation is not a standard continuity equation—it is a *local* evolution equation in which each boundary point responds independently to the local flux $\sigma(x,t)$. Spatial coupling between adjacent points is mediated indirectly through the geodesic equation (E1) and the flux definition (E3), which determine how $\sigma(x,t)$ varies along the boundary. This "spatially decoupled" structure is a consequence of the holographic principle: the boundary is the fundamental stage of the dynamics, and spatial correlations on the boundary are encoded in the flux field rather than in gradient terms. The factor $(1-\rho/\rho_{\max})$ encodes the approach to the Bekenstein bound: as the boundary approaches holographic saturation, the filling rate decreases linearly. This equation is **[Gap 2]**. The linear saturation form is the minimal ansatz compatible with the idempotence requirement $\mu \circ \mu = \mu$; its specific functional form awaits derivation from deeper principles.

Equation (E5) is the mass emergence relation. Mass is not a volume integral but a *boundary* integral—a holographic quantity. Here $\gamma^{(0)}$ is the induced metric on the two-dimensional boundary S inherited from the background spacetime metric $g^{(0)}_{\mu\nu}$, and $\sqrt{\gamma^{(0)}}\,d^2x$ is the invariant area element. The factor $h\nu/c^2$ converts the information count to equivalent mass via the Einstein relation $E = h\nu = mc^2$. Dimensional consistency is verified: ρ has units $[L^{-2}]$ (information per unit area), $\sqrt{\gamma^{(0)}}\,d^2x$ has units $[L^2]$, their product is dimensionless (total information count), $h\nu/c^2$ has units $[M]$, yielding m in units of $[M]$.

Equation (E6) is the Bekenstein bound [4]—a universal upper limit on information density imposed by black hole thermodynamics. In its standard form, the Bekenstein bound states that the entropy S of a bounded system of radius R and energy E satisfies $S \le 2\pi k_B RE/(\hbar c)$. Converting to information $I = S/(k_B \ln 2)$ and expressing the bound as a surface density on a spherical boundary $\rho_{\max} = I/(4\pi R^2)$ yields $\rho_{\max} = 1/(4\ell_P^2 \ln 2)$, where $\ell_P = \sqrt{G\hbar/c^3}$ is the Planck length. This bound provides a natural saturation scale that prevents runaway accumulation of mass in (E5).

5.2 Holographic Boundary as Mathematical Necessity

The six-equation skeleton imposes a mathematical requirement independent of specific physical hypotheses: the information density on any closed boundary is bounded by the Bekenstein limit $\rho_{\max} = 1/(4\ell_P^2 \ln 2)$. When applied with the Hubble horizon as the integration surface, the mass emergence equation (E5) yields a definite mathematical consequence:

$$\boxed{M_{\text{DM}} = \frac{\pi c^3}{G H_0 \ln 2}} \tag{15}$$

This expression contains no free parameters—it is a direct mathematical consequence of applying the six-equation skeleton to the cosmological horizon, structurally analogous to the holographic principle [5, 9]. A heuristic derivation sketch is provided below; a rigorous derivation is reserved for a subsequent publication.

Derivation Sketch

The Hubble horizon is a spherical surface of radius $R_H = c/H_0$. Its surface area is $A_H = 4\pi R_H^2 = 4\pi c^2/H_0^2$. On cosmological timescales, the information density on this horizon is driven to saturation: $\rho \to \rho_{\max} = 1/(4\ell_P^2 \ln 2)$. The total information on the horizon is then $I = \rho_{\max} A_H = 4\pi c^2/(4\ell_P^2 H_0^2 \ln 2)$. Using $\ell_P^2 = G\hbar/c^3$, this becomes $I = \pi c^5/(G\hbar H_0^2 \ln 2)$. Converting information to mass via (E5)—each bit of information corresponds to one energy quantum $h\nu$, $m = h\nu/c^2$—gives $M = (h\nu/c^2)I$. The characteristic frequency is set by the Hubble scale: $\nu \sim H_0$, yielding $M \sim (hH_0/c^2) \cdot \pi c^5/(G\hbar H_0^2 \ln 2) = \pi c^3/(GH_0 \ln 2)$. The cancellation of $\hbar$ is noteworthy: the final expression depends only on classical constants and the geometric factor $\pi/\ln 2$.

This derivation is heuristic. The identification of the characteristic energy quantum scale $\nu \sim H_0$ is dimensional analysis rather than a rigorous result; treating the Hubble horizon as a static surface neglects its dynamical evolution; the direct substitution of the saturation density $\rho_{\max}$ assumes that the cosmological horizon has already reached the Bekenstein limit, which requires independent justification. The precise identification of the energy quantum scale ν in cosmological contexts is labeled [**Gap 12**].

The theoretical significance of this result lies not in numerical matching but in its *structural implication*: it shows that the holographic boundary is not an optional feature of the framework but a mathematical necessity. The Bekenstein bound (E6) serves as a saturation limit that prevents runaway accumulation of mass, and the Hubble horizon is the natural integration surface for mass emergence at cosmological scales. Whether this structural implication corresponds to physical reality is a question for experimental testing (§6, Gap 7).

5.3 Gauge Groups as Mathematical Constraints

The Reproduction paradigm, through its p-adic mathematical structure, imposes a specific mathematical constraint on admissible gauge groups. This is not a phenomenological compatibility requirement but a structural necessity arising from the internal logic of the framework:

- **Prime-group correspondence:** The classification theorem for p-compact groups [18] establishes a one-to-one correspondence between primes p and admissible gauge group structures. In CNT, the primes appearing in the fission of reproduction events are not arbitrary—they are determined by reproductive history. The mathematical requirement is that the gauge group of any CNT-compatible physical theory must be realizable within the p-compact classification at the primes marked by reproductive history.

- **Standard Model as a test case:** The Standard Model gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$ suggests a natural correspondence with the fission primes $p = 3$ (triple fission) and $p = 2$ (double fission)—as one illustrative example, not a unique assignment. This correspondence is a structural analogy, not a derivation—it is labeled [**Gap 5**].

- **Continuum limit requirement:** In the limit of vanishing 4-simplex edge length $\ell_0 \to 0$, the discrete p-adic structure must recover the continuous gauge field theory of the Standard Model. This is a mathematical self-consistency requirement, not a phenomenological prediction—labeled [**Gap 4**].

- **General relativity limit:** In the classical limit $\hbar \to 0$ of coarse-grained discrete geometry, the framework must recover the Einstein field equations. These equations are already incorporated in the (E1)–(E5) feedback loop, but a rigorous derivation of the limit is [**Gap 3**].

These requirements are mathematical self-consistency conditions that the framework must satisfy to be viable. They are currently open problems (§6), which is a measure of the framework's incompleteness rather than a structural contradiction.

5.4 Gravitational Constant G: Energy Quantum–Closed Nucleus Binding

The gravitational constant G is the *binding coupling constant* between energy quanta and closed nuclei. Both entities respond directly to spacetime curvature independently: closed nuclei *are* curvature excitations, and energy quanta follow geodesics of the background. G measures the binding strength—how readily energy quanta are absorbed and bound within a closed nucleus. The classical stress-energy tensor $T_{\mu\nu}$ is the effective description of this binding; as shown in §2.4, it then aligns with the curvature of the closed nucleus that performs the binding through the Einstein field equations. In CNT, G is not fundamental—it is expected to emerge from the geometric parameters of the binding process. Dimensional analysis suggests:

$$G \sim f(\ell_0, \nu, c, \hbar) \tag{16}$$

where ℓ_0 is the 4-simplex edge scale and ν is the energy quantum frequency. The precise functional form has not yet been derived. This is [**Gap 1**] (§6.1).

6 Open Problems and Gaps

This section must be read as an organic part of the research programme, not as an appendix of secondary issues. The gaps catalogued below are *structural*—their resolution is necessary for the framework to advance beyond its current exploratory stage. Each gap is annotated by type, severity, and current research direction.

6.1 [Gap 1] Derivation of G from First Principles

- **Type:** Mathematical/Physical.

- **Severity:** Critical. The binding coupling constant G currently appears as an external input in the Einstein field equations. For CNT to be a self-contained framework, G must be derivable from the binding dynamics between energy quanta and closed nuclei (ℓ_0, ν, c, $\hbar$).

- **Current direction:** Dimensional analysis constrains the form to $G \sim \ell_0^2 c^3/\hbar$ or similar combinations. The specific functional dependence on the binding geometry and network topology is under investigation.
- **Impact if unresolved:** The framework will remain incomplete—the binding strength of energy quanta to closed nuclei (and thereby the strength of gravity) will not be derived from first principles.

6.2 [Gap 2] Derivation of the Information Density Evolution Equation

- **Type:** Physical.
- **Severity:** High. Equation (E4) $\partial_t \rho = \sigma(1 - \rho/\rho_{\max})$ is a physically motivated ansatz. The choice of linear saturation form is made for its simplicity and connection to the Bekenstein bound, but it has not been derived from deeper principles.
- **Current direction:** Investigation underway into whether (E4) can be derived from the holographic principle combined with the second law of thermodynamics. The linear form may be a first-order approximation to more fundamental nonlinear dynamics.
- **Impact if unresolved:** Quantitative applications that depend on the specific form of (E4) will carry systematic uncertainty. The saturation limit ($\rho \to \rho_{\max}$) is independent of the functional form of the approach to saturation.

6.3 [Gap 3] Rigorous Continuum Limit

- **Type:** Mathematical.
- **Severity:** High. The framework posits that general relativity and quantum field theory emerge as continuum limits of the discrete 4-simplex structure. A rigorous proof that these limits exist and recover the correct effective theories has not been completed.
- **Current direction:** Regge calculus [6] provides a well-established framework for the continuum limit of discrete geometry. The 4-simplex is a Regge cell, and the limit $\ell_0 \to 0$ should recover smooth Riemannian geometry. The quantum field theory limit requires separate treatment.
- **Impact if unresolved:** The framework's claim to recover known physics in appropriate limits will remain unsubstantiated.

6.4 [Gap 4] Quantum Field Theory Limit

- **Type:** Mathematical/Physical.
- **Severity:** High. The framework must demonstrate that the Standard Model effective field theory, including its renormalizable interactions and symmetry structure, emerges from closed nucleus dynamics in the continuum limit.

- **Current direction:** The p-adic structure provides a natural framework for discussing renormalization (see §4.6), but an explicit construction of the QFT limit from discrete dynamics has not been achieved.
- **Impact if unresolved:** The framework will lack a bridge to the most precisely tested physical theory in history.

6.5 [Gap 5] Derivation of Gauge Groups from Reproductive History

- **Type:** Physical/Mathematical.
- **Severity:** High. The hypothesis that the Standard Model gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$ represents distinct structural layers of the same reproduction structure remains a structural analogy rather than a derivation.
- **Current direction:** The classification of p-compact groups [18] provides a mathematical framework for associating primes with gauge groups. The specific correspondence $p = 3 \leftrightarrow SU(3)$ and $p = 2 \leftrightarrow SU(2)$ needs to be established beyond numerical coincidence.
- **Impact if unresolved:** The framework will fail to account for the most conspicuous structural feature of the Standard Model.

6.6 [Gap 6] Mathematization of Proliferation

- **Type:** Mathematical.
- **Severity:** High. The proliferation mechanism—the production of new closed nuclei from existing ones—has been described qualitatively but has not yet received rigorous mathematical formulation. The proliferation algebra, fission conditions, and probability distribution of fission products have not been formalized.
- **Current direction:** Analogies with string splitting (Neumann coefficient structure) and branching processes in probability theory are being explored. The p-adic valuation of reproduction counts provides a natural discrete parameter for the fission condition.
- **Impact if unresolved:** The strong construction of CNT (proliferation, fission, irreducibility, historical dependence) will remain partially undefined.

6.7 [Gap 7] Experimental Test Design

- **Type:** Experimental.
- **Severity:** Critical. At present, the framework has no direct experimental test that distinguishes it from Standard Model + ΛCDM. The holographic boundary consequence (§5.2) yields a mass scale consistent with observation but not a *differential* test.

- **Current direction:** Potential differential signals include: (a) deviations in the small-scale ΛCDM matter power spectrum due to discrete structure; (b) a specific pattern of primordial gravitational waves generated by early-universe reproduction dynamics; (c) cosmic microwave background anomalies related to the discrete structure of spacetime excitations.
- **Impact if unresolved:** The framework will remain in the realm of theoretical speculation without empirical grounding.

6.8 [Gap 8] Initial Conditions and Early Universe

- **Type:** Physical/Cosmological.
- **Severity:** Medium. The framework currently does not specify the initial conditions from which the first closed nuclei emerge. The "origin" of the first closed nucleus—the primordial spacetime excitation—is not addressed.
- **Current direction:** The "big bounce" scenario of loop quantum cosmology, combined with the Reproduction paradigm's concept of maximal reproduction count, suggests a cyclic cosmology. This is speculative.
- **Impact if unresolved:** The framework's cosmological narrative will be incomplete.

6.9 [Gap 9] The Cosmological Constant

- **Type:** Physical/Cosmological.
- **Severity:** Medium. The framework does not address the cosmological constant problem. The vacuum energy density driving the accelerated expansion of the universe is not accounted for by the current version of CNT.
- **Current direction:** The holographic boundary structure may provide a natural regularization of vacuum energy, but this has not been developed.
- **Impact if unresolved:** The framework will fail to address one of the most important problems in fundamental physics.

6.10 [Gap 10] Mathematization of Structural Tension

- **Type:** Mathematical/Physical.
- **Severity:** High. The first link of the dialectics of nature—structural tension—has been characterized qualitatively as inconsistency in physical structures whose mathematical signature is non-zero modular remainder (the irreducible part-whole asymmetry), with its mathematization identified as prime-number incommensurability and modular arithmetic. This has not yet been developed into a fully general mathematical construction.
- **Current direction:** Possible mathematical entry points include: (a) primes p marking incommensurable fission patterns, with modular arithmetic $x \bmod p$ yielding structural-tension quantum numbers—the bidirectional remainder structure of compositional p-adic numbers (see §4.3) provides a concrete instance; (b) correspondence between the

algebraic structure of residue class rings $\mathbb{Z}/p\mathbb{Z}$ and conserved quantum numbers; (c) the p-adic valuation v_p measuring the hierarchical depth of tension. These directions are exploratory.

- **Impact if unresolved:** The first link of the three-link dialectics-of-nature sequence will remain unformalized; while quantity-to-quality (p-adic numbers) and hierarchical emergence (RG) have mathematical anchors, the starting point of the sequence will lack operational definition.

6.11 [Gap 11] Construction of the Historical Path Integral

- **Type:** Math/Physics.
- **Severity:** High. The Historical Path Integral (HPI, §4.7)—a p-adic path integral that restores the discrete historical structure obliterated by the continuum limit of standard QFT—has been stated conceptually but not rigorously constructed. The p-adic Haar integration measure, the precise form of the p-adic corrected action, and the proof of the $p \to \infty$ limit (recovering standard QFT) all remain open.
- **Current direction:** Development of p-adic functional integration on the space of reproductive histories $\mathbb{Z}_p$; construction of the p-adic analogue of the Wiener measure for the HPI; derivation of the $p \to \infty$ limit via the adelic product formula $\prod_p |x|_p \cdot |x|_\infty = 1$.
- **Impact if unresolved:** The "computational hub" problem (§3.6) will lack a rigorous mathematical foundation; coupling constants cannot be derived from reproductive history beyond heuristic dimensional estimates.

6.12 [Gap 12] Determination of the Cosmological ν Scale

- **Type:** Physical.
- **Severity:** Medium. The characteristic energy quantum scale $\nu \sim H_0$ used in the holographic boundary derivation (§5.2) is dimensional analysis rather than a rigorous result. The treatment of the Hubble horizon as a static surface and the direct substitution of the saturation density both require independent justification.
- **Current direction:** Deriving the precise value of ν from the reproduction process and early-universe conditions; determining the effective energy quantum frequency at the Hubble horizon from closed nucleus network dynamics.
- **Impact if unresolved:** The numerical estimate of M_{DM} will carry systematic uncertainty; the precision of this result depends on a rigorous derivation of ν.

6.13 [Gap 13] Architecture of the Closed Nucleus Network

- **Type:** Math/Physics.
- **Severity:** Medium. The working hypothesis of §4.8—that the proton is a network of Planck-scale closed nuclei rather than a single closed nucleus—provides a concrete computational entry point for the p-adic path integral, but is entirely unverified. The

network architecture, the number of evolutionary stages, the mechanism of structural inheritance, and the identification of network components with quark types are all open to revision.

- **Current direction:** Systematic trial of multiple candidate architectures; development of p-adic network models that encode stage-wise evolution through the projective limit structure; comparison of structural predictions (evolutionary layer inheritance, gauge group projection order) with observed gauge structures.
- **Impact if unresolved:** The "computational hub" problem (§3.6) remains without a concrete computational entry point; coupling constant derivation from reproductive history cannot proceed beyond formal statement.

6.14 Gap Severity Summary

Gap	Severity	Type
Gap 1: Derivation of G	Critical	Math/Physics
Gap 2: Derivation of (E4)	High	Physics
Gap 3: Continuum limit	High	Math
Gap 4: QFT limit	High	Math/Physics
Gap 5: Gauge groups	High	Physics/Math
Gap 6: Proliferation	High	Math
Gap 7: Experimental test	Critical	Experiment
Gap 8: Initial conditions	Medium	Cosmology
Gap 9: Λ problem	Medium	Cosmology
Gap 10: Structural tension	High	Math/Physics
Gap 11: Historical Path Integral	High	Math/Physics
Gap 12: Cosmological ν scale	Medium	Physics
Gap 13: Closed nucleus network architecture	Medium	Math/Physics

7 Conclusion

7.1 Summary of the Research Programme

Closed Nucleus Theory is a research programme within Reproduction Physics—a framework that inverts the standard ontological ordering of mass, gravity, and spacetime. Its core achievements to date are:

1. **Conceptual unification:** The framework provides a common ontological origin for mass, gravity, and spacetime structure. Mass is not a source but an emergent readout; gravity is a force fundamentally different from the gauge forces—an ontological response; closed nuclei are not particles but spacetime excitations.
2. **Mathematical structure:** The regular 4-simplex provides the minimal geometric realization, and p-adic numbers are identified through a chain of logical necessity as the necessary mathematical direction of Reproduction Physics—their specific form (e.g., compositional p-adic numbers) is an active research frontier. The six-equation skeleton delineates the mathematical self-consistency conditions that a physical realization must satisfy.

3. **Structural constraints:** The framework identifies mathematical necessities—the Bekenstein bound as a saturation limit, gauge groups as distinct structural layers of the same reproduction structure, and the gravitational constant G as the energy quantum–closed nucleus binding coupling constant—that delineate the research frontier.

4. **Explicit gap catalogue:** All open problems are explicitly labeled and graded by severity, enabling the community to assess the framework's current state of development.

7.2 Methodological Distinction from Closed Self-Consistent Theories

CNT must be clearly distinguished from theories constructed as closed, self-consistent systems from arbitrary axioms. This distinction is methodological, not rhetorical:

- **Starting point:** CNT does not begin from axioms chosen for their explanatory power. It begins from a structural observation about general relativity—the ontological ambiguity in the Einstein field equations—and follows the consequences of resolving that ambiguity in a specific direction. The framework's validity is therefore not a question of internal consistency but of whether the initial ontological judgment is correct.
- **Relationship to existing physics:** A closed self-consistent theory can, in principle, coexist with existing physics by carving out a separate domain. CNT cannot: the curvature-first ontology requires that all phenomena currently described by general relativity and the Standard Model be derivable, in appropriate limits, from the framework's deeper principles. Failure to recover general relativity or quantum field theory in the continuum limit would falsify the framework, not merely leave it incomplete.
- **Falsifiability structure:** A closed self-consistent theory is falsifiable only at its periphery—through empirical tests of its predictions. CNT is falsifiable at its *core*: if the ontological inversion is wrong, the framework has no empirical content and no independent justification. This radical falsifiability is not a weakness but the framework's primary epistemic virtue—it eliminates the possibility of endless ad hoc adjustments.
- **Methodological self-awareness:** CNT explicitly distinguishes what follows from the ontological inversion (hard core), what follows from CNT-specific strong construction choices (protective belt), and what is conjectural extrapolation (gaps). This stratified structure prevents confusion between the framework's foundational commitments and its current speculative extensions.

This distinction is not an incidental clarification. In an era where AI-assisted theory construction can generate formally self-consistent frameworks that "explain" selected phenomena while remaining empirically disconnected, methodological self-awareness—explicitly stating what a theory *requires* to be true, what it *assumes*, and under what conditions it *fails*—is the minimum condition for a research programme to be taken seriously.

7.3 The Single-Point Falsifiability Condition

The entire CNT framework rests on a single claim: that curvature (spacetime geometry) is ontologically prior to mass-energy (the stress-energy tensor). This is not a claim about dynamics or empirical fitting—it is an ontological claim about *what exists prior to what* in spacetime structure.

The condition is binary:

- If the curvature-first ontology is **correct**, then CNT provides the correct conceptual starting point for understanding mass, gravity, and spacetime structure. The chain from curvature through geodesic binding, reproduction, p-adic encoding, and holographic boundary emergence unfolds as a path of logical necessity, once the core principles are accepted. The specific mathematical constructions (4-simplex, six-equation skeleton, proliferation algebra) are then natural tools—among other possible choices—for traversing this path.

- If the curvature-first ontology is **wrong**—if mass-energy is ontologically prior to, or on par with, geometric structure—then the entire CNT framework collapses. There is no weakened version, no phenomenological residue, no tunable parameter that could rescue it. The theory is designed to have no fallback position, because a theory that can survive the falsification of its foundational claim is not a theory but a rhetorical posture.

This condition is not imposed by the researcher's preferences. It is a structural consequence of the framework's architecture: the ontological inversion is the root of the entire concept chain (§3.5). Sever the root and the tree dies. This makes CNT maximally fragile and maximally honest—precisely the qualities that distinguish a genuine research programme from an exercise in theoretical aesthetics.

7.4 Invitation to the Community

This paper is an invitation to a research programme, not a declaration of correctness. The CNT framework is submitted to the physics community for scrutiny, criticism, and—should it be judged to have merit—collaborative development. The appropriate response to a research programme is not belief but *critical engagement*:

- **To theorists:** Are the logical inferences in the concept chain (§3.5) valid? Do alternative mathematical structures exist that could encode the same physical content? Do the gaps (§6) admit solutions within the framework, or do they indicate structural defects?

- **To phenomenologists:** Can the six-equation system be extended to make differential predictions? Do existing datasets exist that could constrain the framework's parameters?

- **To mathematicians:** Can the proliferation algebra be formalized? Can the continuum limit be rigorously proved? Is the p-adic structure truly necessary, or do alternatives exist?

- **To philosophers of physics:** Is the ontological inversion coherent? Can the concept of "spacetime excitation" withstand scrutiny? Is the relationship between Reproduction Physics and historical materialism philosophically sound? Does the mathematization scheme of the Aufhebung-elevated dialectics of nature (structural tension $\rightarrow$ quantity-to-quality $\rightarrow$ hierarchical emergence) capture the physical kernel of dialectics?

The framework makes claims precise enough to be tested against both logical consistency and empirical observation. The author welcomes the identification of errors, for error is the engine of progress.

7.5 Final Word

Reproduction Physics begins from a single structural observation: the Einstein field equations encode an ontological ambiguity between curvature and stress-energy. Resolving this ambiguity in favor of curvature generates a chain of logical necessity—from intrinsic geometry through geodesic binding, reproduction, p-adic numbers, holographic boundaries, to gauge groups, and—should the framework be correct—to the dark matter that fills the universe. Whether this chain is a genuine path to deeper understanding or an elaborate error is for the community to judge. The author welcomes that judgment, whatever its outcome, for in theoretical physics the only true failure is failing even to be wrong.

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